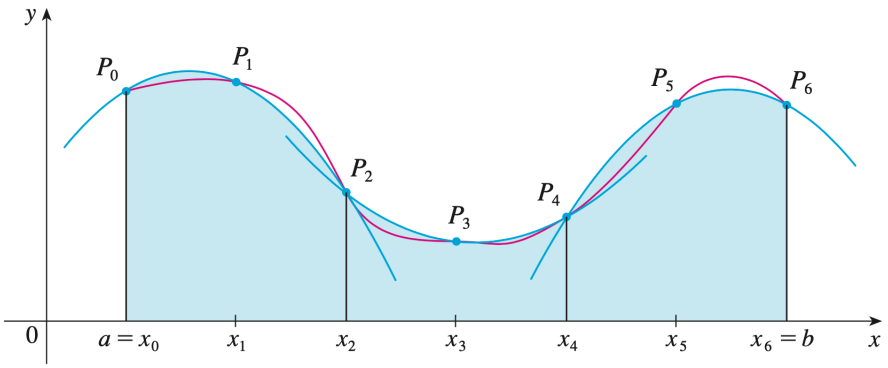
1 Simpson's Rule

With the midpoint rule, we estimated areas of regions under curves by using rectangles. In a sense, we approximated the curve with piecewise constant functions. With the trapezoidal rule, we approximated the curve by using piecewise linear functions. We now consider approximating the curve using piecewise quadratic functions. This is known as *Simpson's Rule*, named after English mathematician Thomas Simpson (1710-1761). [1]

(i) This time we partition the interval $[a,b]$ into an *even* number of subintervals of equal width $\frac{b-a}{n}$.

(ii) Over the first pair of subintervals we approximate $\int_{x_0}^{x_2} f(x)dx$ by $\int_{x_0}^{x_2} p(x)dx$, where $p(x) = Ax^2+Bx+C$ is the quadratic polynomial passing through $(x_0, f(x_0)), (x_1, f(x_1))$, and $(x_2, f(x_2))$.

(iii) Over the next pair of subintervals we approximate $\int_{x_2}^{x_4} f(x)dx$ with the integral of another quadratic function, which passes through $(x_2, f(x_2)), (x_3, f(x_3))$, and $(x_4, f(x_4))$, and so on.



[1] Not to be confused with nuclear safety inspector Homer Simpson.

1. Show that $\int_{x_0}^{x_2} p(x)dx = \int_{x_0}^{x_2} (Ax^2 + Bx + C)dx$ can be written as

$$\frac{x_2 - x_0}{6}(2A(x_2^2 + x_0x_2 + x_0^2) + 3B(x_2 + x_0) + 6C).$$

(*Hint*: Use the difference of squares and difference of cubes factorization formulas.)

2. Show that your answer to Problem 1 can be written as

$$\frac{\Delta x}{3}((Ax_2^2 + Bx_2 + C) + (Ax_0^2 + Bx_0 + C) + A(x_0 + x_2)^2 + 2B(x_2 + x_0) + 4C).$$

Since $p(x)$ passes through the points$(x_0, f(x_0)), (x_1, f(x_1))$, and $(x_2, f(x_2))$, we have

$$f(x_0) = p(x_0) = Ax_0^2 + Bx_0 + C$$
$$f(x_1) = p(x_1) = Ax_1^2 + Bx_1 + C$$
$$f(x_2) = p(x_2) = Ax_2^2 + Bx_2 + C.$$

Note also that $x_2 - x_0 = 2\Delta x = \frac{b-a}{n}$, and $x_2 + x_0 = 2x_1$, since $x_1 = \frac{x_2+x_0}{2}$.

3. Use the above formulas to show that your answer to Problem 2 can be written as

$$\frac{\Delta x}{3}(f(x_2) + 4f(x_1) + f(x_0)).$$

4. We have just shown that $\int_{x_0}^{x_2} f(x)dx \approx \frac{\Delta x}{3}(f(x_2)+4f(x_1)+f(x_0))$. If we approximate $\int_{x_2}^{x_4} f(x)dx$ with the same method, then we should get

$$\int_{x_2}^{x_4} f(x)dx \approx _____.$$

5. Therefore,

$$\int_{x_0}^{x_4} f(x)dx \approx _____.$$

6. Continuing in this way, we find that (fill in the coefficients)

$$\int_{x_0}^{x_n} f(x)dx \approx \frac{\Delta x}{3}(_f(x_0)+_f(x_1)+_f(x_2)+_f(x_3)+_f(x_4)+\cdots+_f(x_{n-2})+_f(x_{n-1})+_f(x_n)).$$

2 Error bound

Theorem 2.1. *Let $f(x)$ be a continuous function over $[a,b]$ having a fourth derivative $f^{(4)}(x)$ over this interval. If $M = \max\{|f^{(4)}(x)|\}_{x\in[a,b]}$, then the upper bound for the error in approximating $\int_a^b f(x)dx$ using Simpson's Rule is*

$$E_n^S \leq \frac{M(b-a)^5}{180n^4}.$$

As before, we will postpone the proof and move on to consider examples.

3 Exercises

8. Use Simpson's rule with $n = 2$ to estimate $\int_0^1 x^3 dx$ and find an upper bound for the error in your approximation.

9. Use Simpson's rule with $n = 10$ to estimate $\int_0^1 e^{x^2}\,dx$ and find an upper bound for the error in your approximation.

10. What value of n should be used to guarantee that using Simpson's Rule to estimate $\int_0^1 e^{x^2} dx$ is accurate to within 0.01?

4 Relationship between Simpson's Rule and Midpoint, Trapezoidal Rules

11. Prove that $S_n = \frac{2}{3}M_{\frac{n}{2}} + \frac{1}{3}T_{\frac{n}{2}}$. (*Hint*: Using a partition P of n (with n even) intervals of width Δx to calculate S_n, the partition used for $M_{\frac{n}{2}}$ and $T_{\frac{n}{2}}$ has width $2\Delta x$.)